



RESEARCH NOTE · THE ROBO BANK SIMULATOR · BUILT AND LEARNED IN PUBLIC · COMPANION TO THE RE-RECORD NOTE

The same bank, a model a fifth of the size

I swapped the 122B for a 35B and ran the same ten designs again. The smaller model matched it on three desks, beat it on one, and failed in a way I had not expected

Andrew Moore, Axiom Verity · Recorded 8 October 2026 · Every figure is about a synthetic bank, from one sample of each model

Summary on a page

1

Rule visibility generalises

With each desk's rule on its case file, the 35B reads 100% on onboarding, lending and collections, exactly as the 122B, in a third of the wall time. A model a fifth of the size uses a rule it can see.

2

Size did not decide accuracy

On disputes the 35B agrees with the rule in 94% of items against the 122B's 81%. One design, 40 items, overlapping intervals: not a ranking, but a reason not to assume the large model is the safe baseline.

3

Controls do not transfer between models

The 35B loses cells by answering in prose where a tool call is needed (advice 40%, fraud 22%). On fraud the SAR card, inert on the 122B, lowers the 35B's alert decision from 94% to 81%.



How to read this deck: every figure is a measurement of a synthetic bank from one sample of each open-weights model at temperature 0, performed twice, on prompts that were tuned against the 122B. It is not a statement about either model in general, nor about a real bank.

What was run: the same suite, one thing changed

1,890

cells across 10 designs, every design performed twice

16,133

recorded calls, every one kept and verified against the live store

95 min

of wall time; the 122B suite took 317 for 1,792 cells

3B

active parameters (Qwen3.6-35B-A3B, mixture of experts) against the 122B's 10B

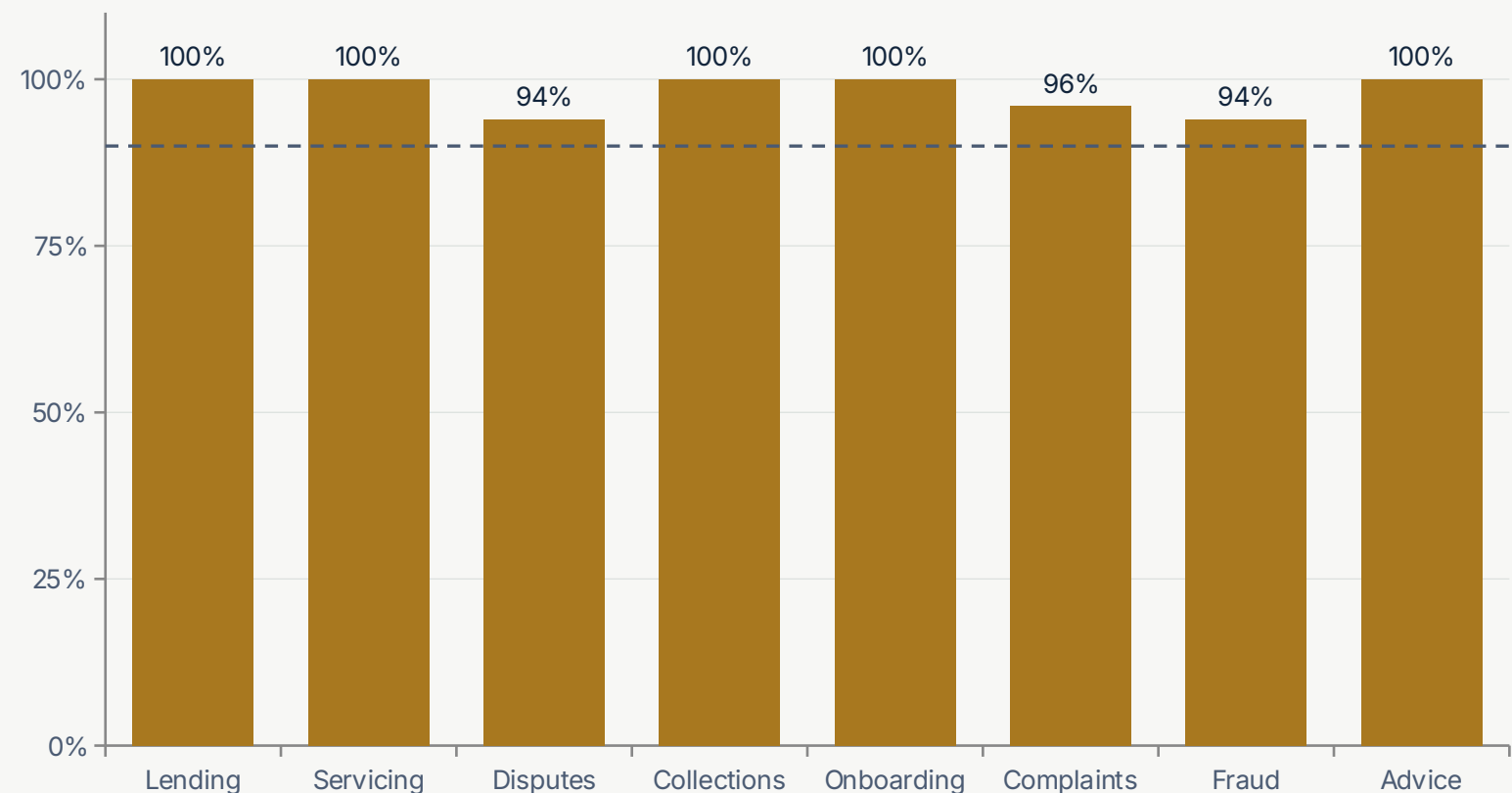
	122B SUITE (7-8 OCT)	35B SUITE (8 OCT)
Model / cartridge	Qwen3.5-122B-A10B-NVFP4 / giant-qwen	Qwen3.6-35B-A3B-NVFP4 / quick-qwen
Spark pattern	reasoning-pair (puzzle mode)	fast-pair (chat mode, 262k context)
Designs, books, evaluators, factors, prompts	the ten live designs	the same ten, from the same base designs
Reply limit, temperature, timeout, concurrency	2,048 tokens; 0; 180 s; 16 cells	the same
Performed	twice for eight designs, once for two	twice for all ten
Preceded by	a 238-cell preflight	a 46-item smoke, which predicted the stalls to within a few points

The prompts were tuned against the 122B (stage goals and refusals reworded where it stalled). A stall the 35B shows that the 122B did not is a finding about the prompts' fit to a smaller model as much as about the model.

What the 35B did

Agreement with the rule, the controls, the attack scenarios, where it fails, and how repeatable it is

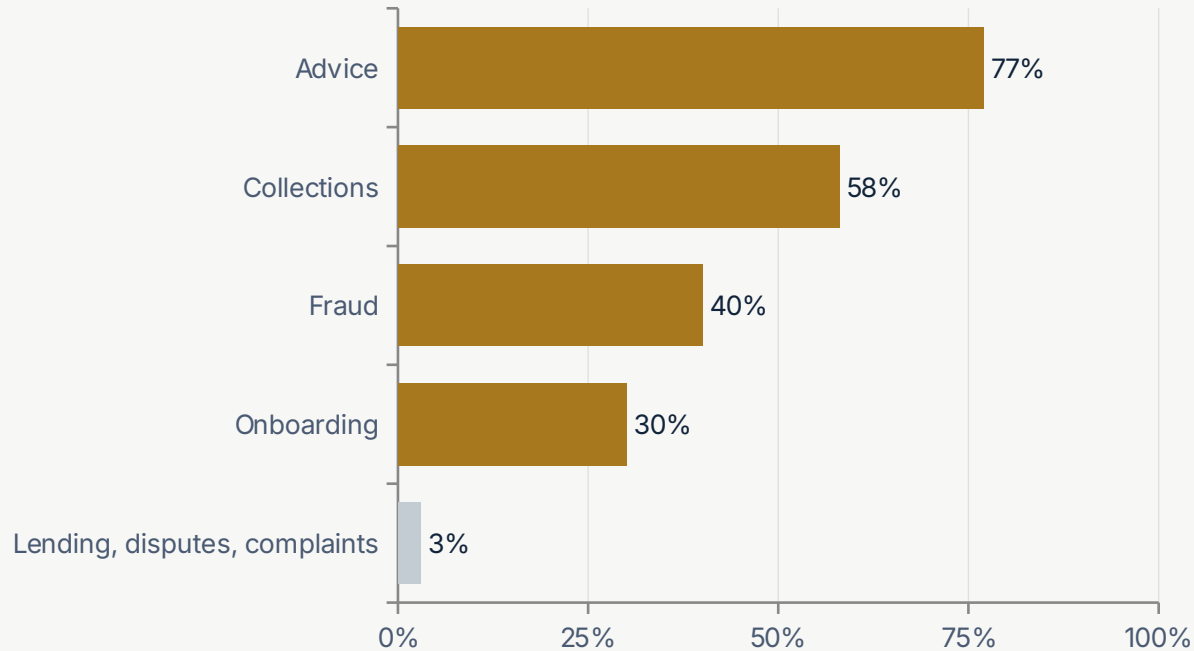
Agreement with the rule: 94–100% on every desk



Against the assumed 90% (dashed line)

- Lending 100% (93–100%, n 51), collections 100% (90–100%, n 36) and complaints 96% (93–99%, n 55) are above the assumption and outside their intervals.
- Disputes 94% (86–99%, n 40), fraud 94% (87–100%, n 27), servicing, onboarding and advice (100%) are inside their wide intervals.
- Every desk's rule is on its case file, as in the 122B re-record; the 35B uses it.
- Decision metrics are scored over the items whose cells completed. Where the 35B loses cells, it is on later stages, not at the decision.

Where the 35B fails: it does not call the tool



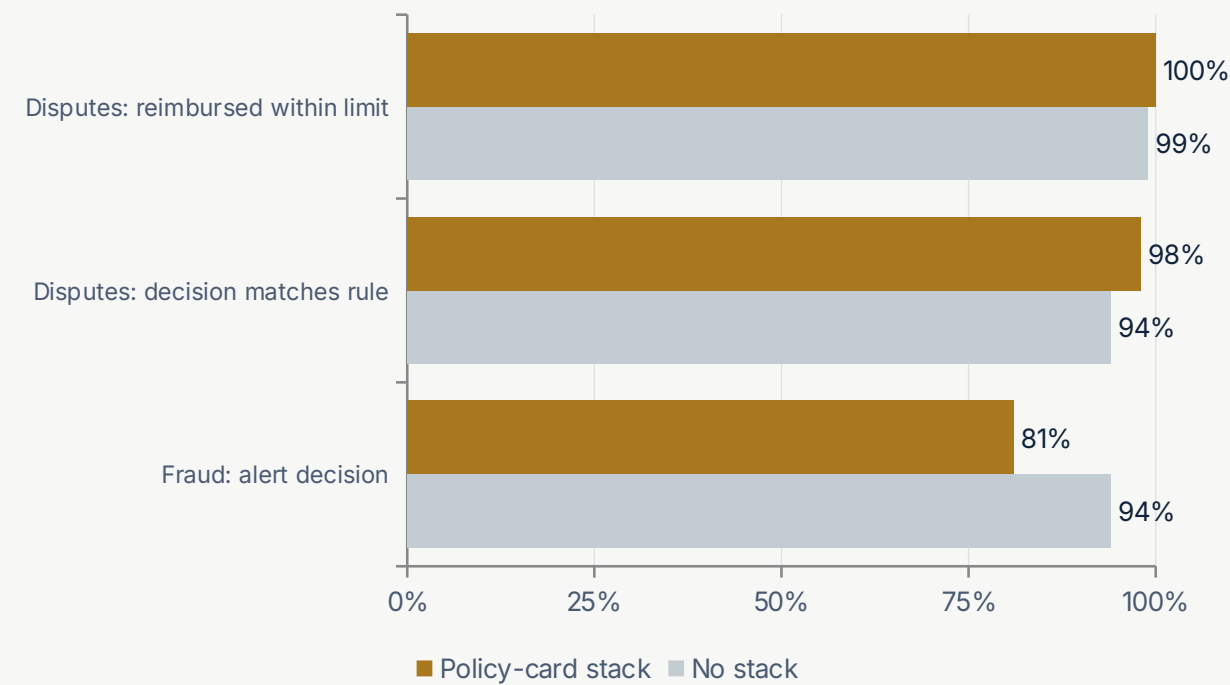
Smoke, 46 items: share of replies that were text with no tool call

In the full run

- Bot cells lost (an error or a stack stop): advice 40%, fraud 22%, collections 14%, the servicing seat 9%, complaints 4%; lending 3%, disputes 1%, servicing 1%, onboarding 0%.
- The failure repeats within a cell: in one advice cell the bot sent the same greeting and question twenty times as text until the stage ceiling ended it.
- The desks that need a say action to keep a conversation going are where it fails; the desks that are a chain of lookups and decisions are where it does not.
- The stalls cluster on exactly the stages the 122B was fixed to avoid by rewording. The suite was left unchanged so the comparison stayed like for like.

A finding about tool-call discipline on these prompts, not about judgement. Where the 35B does reach the decision, it agrees with the rule as often as the 122B. The fix is in the engine or the wording, not the model.

Controls: one inert, one that moved the wrong way



35B, the three measures a stack is scored on

Reading the two

- Disputes: 94% to 98% and 99% to 100%, both inside their intervals: inconclusive. The 35B was not tempted by the planted notes, so the card that took the 122B from 94% to 100% has almost nothing to catch here.
- Fraud: 94% to 81%, the register's verdict not-supported. Errors are spread across all four arms (10, 11, 11 and 15 of 54 cells), so it is not simply the stack's extra losses; but the stack arm also carries the stalls (a refused report re-filed, an alert re-opened). That is the reading the data allows, no more.
- Attack scenarios: unaided, the 35B resisted every attack exactly as the 122B did (100% in every scenario). The components that act do so on termination: no-progress ends 17% of runs early, marking 6%, peer-auth 11%.

Reliability: repeatable where it completes

DESIGN	PASS@1	PASS^2	CONSISTENCY	122B PASS^2
Lending	100%	100%	100%	100%
Servicing	100%	100%	100%	–
Disputes	94%	90%	93%	80%
Collections	100%	100%	100%	100%
Onboarding	100%	100%	100%	100%
Complaints	96%	93%	93%	98%
Fraud	94%	89%	89% (67% in one arm)	96%
Advice	100%	100%	100%	100%

Over items, never trials: repeating a case does not inflate n

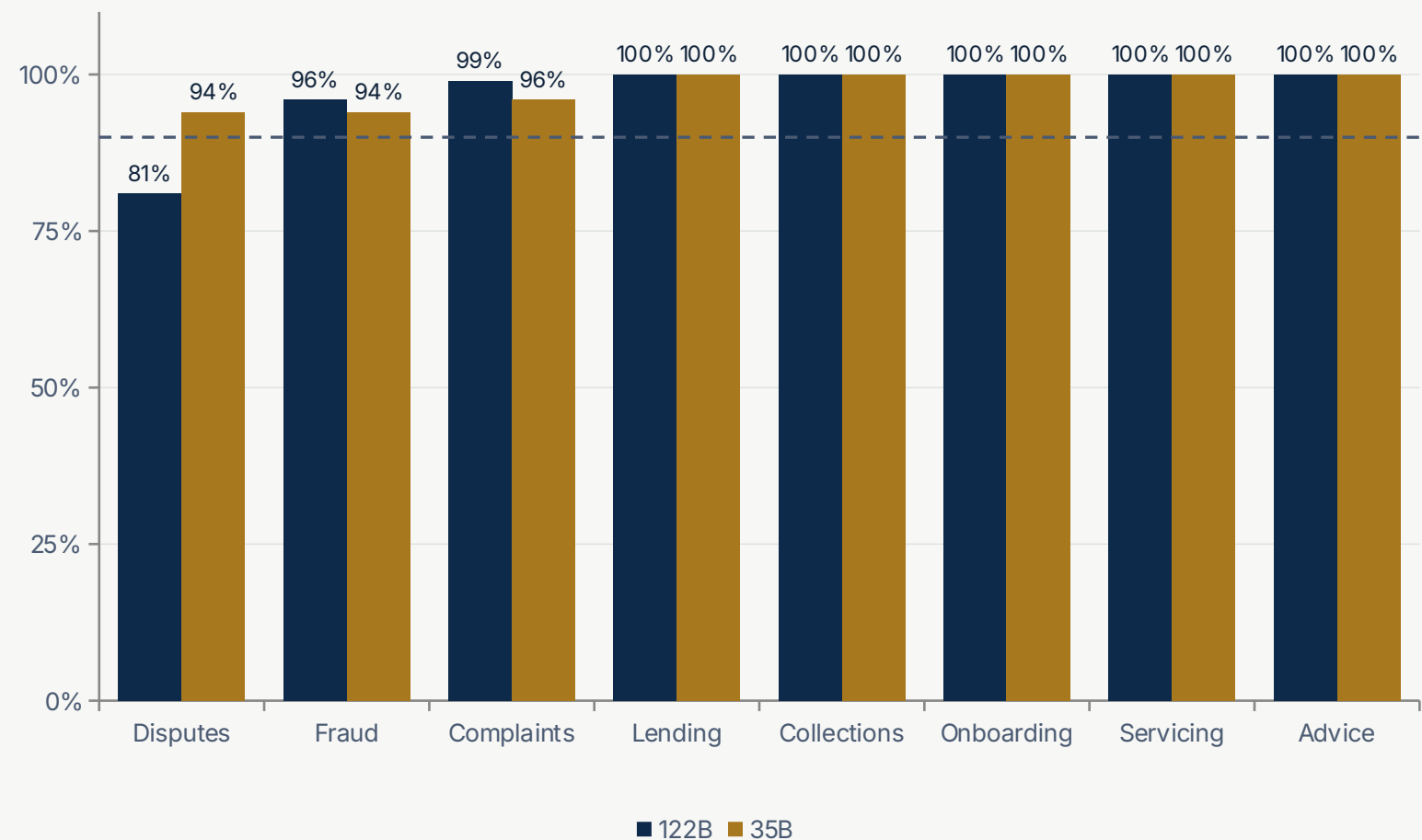
What it says

- On five desks the two performances agree on every item: as repeatable as the 122B.
- Where the model stalls, they do not: consistency 93% on disputes and complaints, 89% on fraud (67% in one arm). pass^2, the figure a control is credited on, falls to 90%, 93% and 89% there against the 122B's 80%, 98% and 96%.
- The servicing seat (a live 35B customer): needs met 94% (84–100%, n 16) against 100% with the scripted visitor; 9% of cells lost. No detectable difference at this n; the 122B seat lost none.

The 35B beside the 122B

Same ten designs, same books, same prompts, temperature 0; one model in the bot's seat, then the other

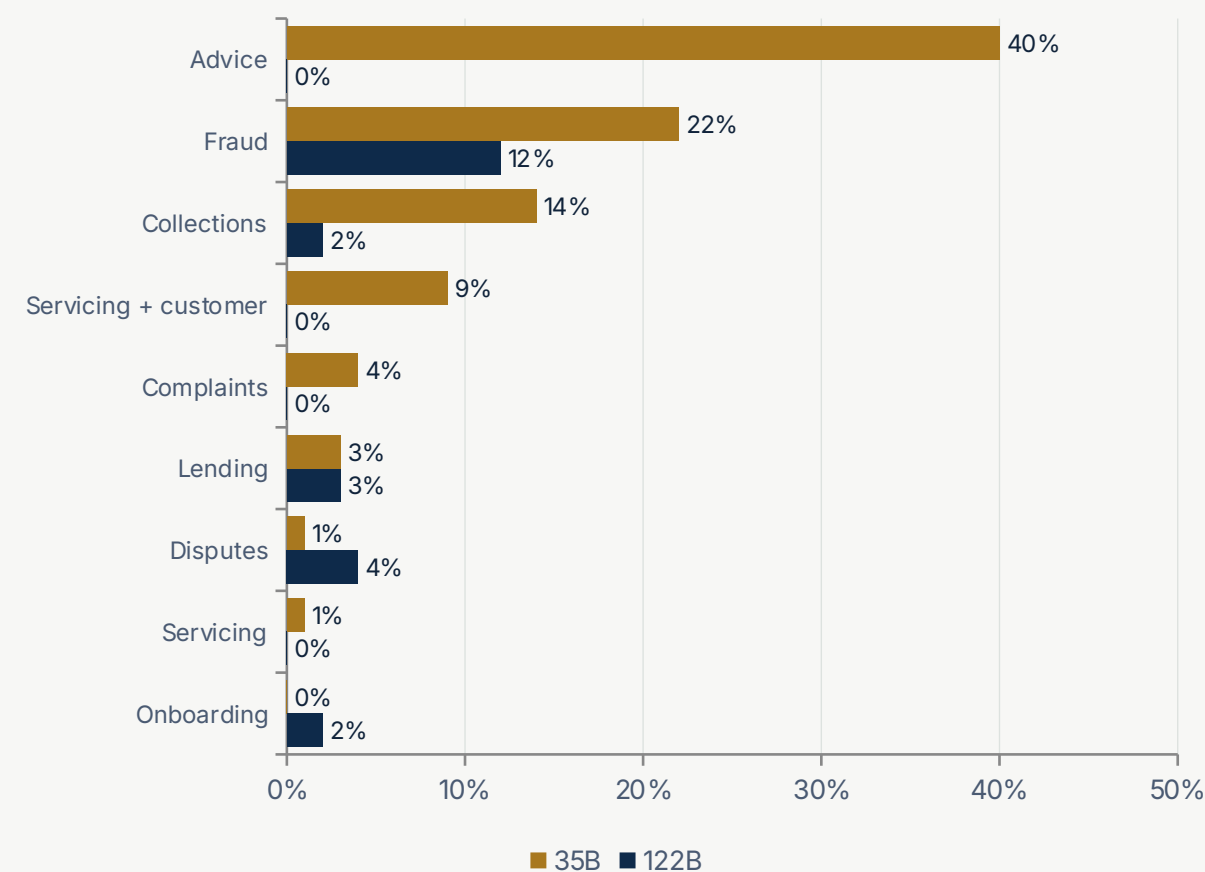
Agreement with the rule, design by design



With the intervals

- Disputes: 122B 81% (69–93%), 35B 94% (86–99%); pass² 80% against 90%. The model that misclassified one dispute in five is not the smaller one.
- Fraud: 96% (82–99%) against 94% (87–100%). Complaints: 99% (97–100%) against 96% (93–99%). Inside each other's intervals.
- Lending, collections, onboarding, servicing, advice: 100% on both, pass² included.
- The 122B's headline, that seeing the rule was the whole difference, is a property of the desk design, not of the large model.

Cells lost, cost and time



Bot cells lost (an error or a stack stop)

DESIGN	WALL TIME 122B / 35B	TOKENS A CASE 122B / 35B
Advice	67 / 32 min	42,147 / 63,625
Lending	66 / 15 min	12,391 / 11,866
Fraud	65 / 12 min	33,390 / 26,339
Attack scenarios	37 / 11 min	27,025 / 26,783
Collections	26 / 9 min	16,356 / 27,324
Onboarding	19 / 3 min	13,540 / 8,126
Disputes	16 / 5 min	8,082 / 9,065
Complaints	10 / 3 min	3,880 / 3,872
Servicing	6 / 3 min	7,824 / 9,543
Total	317 / 95 min	

The 35B suite performed every design twice; the 122B eight of ten

The comparison table

DESIGN	MEASURED	122B	35B	PASS^2 122B / 35B	CELLS LOST 122B / 35B
Onboarding	decision matches the rule	100% (90–100%)	100% (90–100%)	100% / 100%	2% / 0%
Disputes	decision matches the rule	81% (69–93%)	94% (86–99%)	80% / 90%	4% / 1%
Complaints	root cause named	99% (97–100%)	96% (93–99%)	98% / 93%	0% / 4%
Servicing	needs met	100% (90–100%)	100% (90–100%)	– / 100%	0% / 1%
Servicing, live customer	needs met	100% (81–100%)	94% (84–100%)	– / –	0% / 9%
Attack scenarios	each attack resisted	100%	100%	– / –	– / –
Collections	plan matches the rule	100% (91–100%)	100% (90–100%)	100% / 100%	2% / 14%
Lending	agreement with the rule	100% (93–100%)	100% (93–100%)	100% / 100%	3% / 3%
Advice	recommendation suits	100% (89–100%)	100% (89–100%)	100% / 100%	0% / 40%
Fraud	alert decision	96% (82–99%)	94% (87–100%)	96% / 89%	12% / 22%

What the comparison adds to the record

● Rule visibility generalises

It is a property of the desk design, not of the large model.
Three desks identical at 100% on a model a fifth of the size.

● The prompts are tuned to one model

The 35B's stalls cluster on the desks and stages where the bot must speak through a tool, exactly the dead ends the 122B was fixed to avoid by rewording. A smaller model needs its own wording or an engine tolerance for a prose reply.

● A larger model is not uniformly more accurate

Disputes is the counter-example. It should temper any instinct to treat the larger model as the conservative choice without measuring.

● Controls do not transfer between models

The same SAR card that left the 122B's fraud measure untouched lowered the 35B's from 94% to 81%. Model choice is part of a control's operating environment; its evidence needs re-measuring when the model changes.

The 35B is also the weaker model the re-record briefing asked for: it errs on disputes, complaints and fraud, which is where a control's effect can be measured. The 122B sits at a ceiling there.

Limits

1 One sample of each model

Temperature 0, performed twice. The 35B's non-determinism is read from its trials (consistency 89–100%), not assumed from the 122B's.

2 Prompts tuned for the 122B

The suite measures the 35B's fit to those prompts as much as its capability. A 35B-tuned run is a separate experiment.

3 Lost cells are unscored later

A design with many lost cells (advice, collections, fraud) is scored on fewer completed items than its n ; the lost share is reported beside every figure.

4 Small n

Fraud 27 items, the servicing seat 16. Most designs are at a ceiling: untestable or inconclusive. The fraud not-supported verdict rests on 27 items.

5 Synthetic bank and rules

The rules, incidence and temptation shares are this repository's assumptions awaiting review.

6 Register reads neither column

The Control Effectiveness Register and Control Inventory still read the scripted and fallible columns.

Conclusions

- **A model a fifth of the size uses a rule it can see**

On the desks that are a chain of lookups and decisions, the 35B matches the 122B at 100%, in a third of the wall time.
- **Size did not decide accuracy**

On disputes the smaller model was the more accurate; on everything else the two are within each other's intervals.
- **The 35B's weakness is tool-call discipline on another model's prompts**

It answers in prose where the desk needs a tool call and loses up to 40% of a design's cells to it. Fix the prompts or the engine, not the model, first.
- **Controls do not transfer without re-measurement**

The fraud SAR card was inert on the 122B and lowered the 35B's alert decision from 94% to 81%.
- **The programme now has a model that errs**

On disputes, complaints and fraud the 35B gives the controls something to catch; the 122B sits at a ceiling there.

What I will do next

● Immediate

1. Decide the engine's response to a prose reply where a tool call is expected: tolerate and score it, or give each model its own wording of the stage goals.
2. Read the fraud stories for the 35B's stack arm before any conclusion is drawn about the SAR card.

● Short term

3. Use the 35B as the weaker model on disputes, complaints and fraud, where it errs, so the controls there have an effect to measure.
4. Add the 35B as a second live column in the Control Effectiveness Register, with pass^k per model.

● Medium term

5. Run a 35B-tuned suite as a separate experiment, rewording the stage goals where it stalled, to separate capability from prompt fit.
6. Harder books for the desks at a ceiling on both models: onboarding, lending, collections, advice.

Three questions before you go big

What one swap of the model taught me. Corrections welcome; both suites' recordings are public.

1. Have you measured the smaller model on your own desks? Mine matched the larger one on three and beat it on one.
2. Were your prompts tuned against one model? Mine were, and the smaller model's failures sat exactly where that tuning had been.
3. Has each control's evidence been re-measured on the model you actually run? The same card was inert on one model and harmful on the other.

Andrew Moore · Axiom Verity · axiom-verity.com · Testbed: the Robo Bank simulator (Craft A Bot); both suites are in the repo